

PCB Design Process Output Matrix

Current July_04 status based on KiCad 10.0.3 validation after non-netlist manufacturability cleanup.

PCB Step	Generated Output	Package Folder	Current Status
1. Schematic Creation / Validation	Native KiCad schematic stub and readable dual-NCE/TFLN overview.	04_front_end_schematic	Complete schematic/netlist source required to close 214 parity warnings.
2. BOM	Current BOM workbook with grouped and expanded placements.	05_bom	Current July_04 review list; owner signoff required for AVL, quantity, DNP/populate, substitutions, and parity.
3. Footprint Creation / Validation	Project-local footprint libraries and reconciliation evidence.	06_footprints_symbols	82 library/footprint mismatch warnings remain for engineering review.
4. Netlist	Board net table and available reference exports.	07_netlists	Authoritative schematic/netlist source still required; current schematic is a stub.
5. Board Outline / Mechanical Detail	KiCad Edge.Cuts and mechanical notes.	08_board_outline_mechanical	Current Edge.Cuts approx. 305.0 mm x 260.5 mm; chassis constraints need confirmation.
6. Stackup / Constraints Setting	Board setup stackup and manufacturer checklist.	09_stackup_constraints	4-layer High-Tg FR4 review stackup and ENIG finish metadata added.
7. Component Placement	Placement ECO notes and current PCB source.	10_component_placement	Six board-only fiducials added; placement still needs chassis, thermal, and SI review.
8. Routing and Planes	Current routing/plane state and review notes.	11_routing_planes	0 unconnected items in current DRC; real schematic/netlist needed for electrical closure.
9. Silkscreen / DRC / FAB Notes	Silkscreen/fab-note review and DRC records.	12_silkscreen_drc_fab_notes	Silkscreen-over-copper warnings cleared; 82 library warnings remain.
10. Gerber Generation	Review Gerbers, drill, drill map/report, and Gerber index.	13_gerbers_drill	Post-cleanup Gerber/drill review export generated and copied to Step 01 pair.
11. DFA	Assembly/DFA checklist and action matrix.	14_dfa_assembly	Review placement, polarity, DNP/populate, and BOM owner signoff.
12. DFM	DFM/manufacturability checklist and action matrix.	15_dfm_manufacturability	Submit post-cleanup Gerber/drill package for DFM plus stackup and chassis confirmation.